

LAB 1 - NETWORK MODEL

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Subject: DBMS

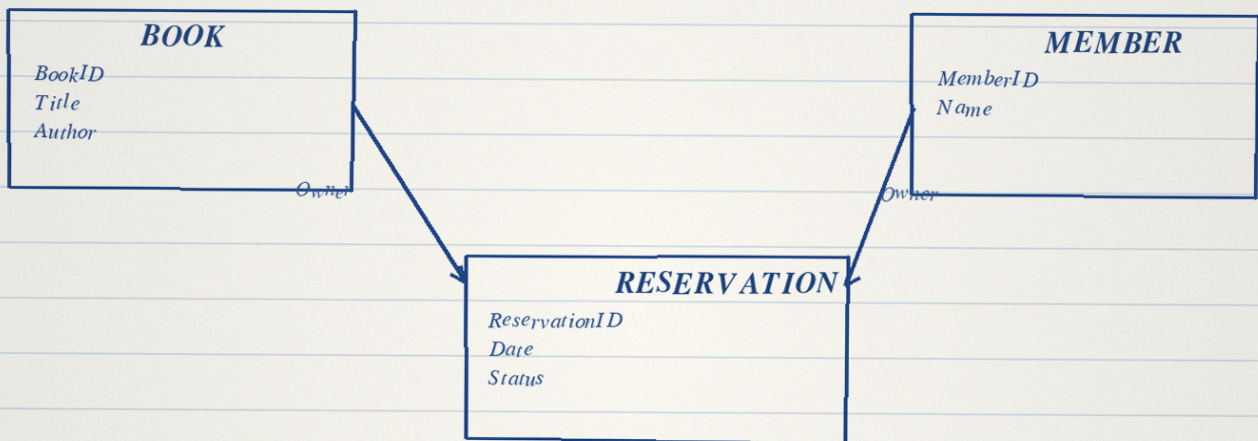
Q1. Network Model

Re-model the same data using a Network (CODASYL-style) structure with owner-member set relationships, resolving the many-to-many case (Book $\leftrightarrow$ Member via Reservation).

Answer:

In the CODASYL Network Model, records are connected using owner-member set relationships. The many-to-many relationship between BOOK and MEMBER is handled by using RESERVATION as a linking record.

Network Structure:



Owner-Member Sets:

1. BOOK_RESERVATION_SET $\rightarrow$ Owner = BOOK, Member = RESERVATION
2. MEMBER_RESERVATION_SET $\rightarrow$ Owner = MEMBER, Member = RESERVATION

How the M:N relationship is resolved:

A book can have many reservations and a member can also have many reservations. Each RESERVATION connects one BOOK with one MEMBER. Example:

R1 : B101 $\rightarrow$ M01

R2 : B101 $\rightarrow$ M02

R3 : B102 $\rightarrow$ M01

Conclusion:

The Network Model handles the Book $\leftrightarrow$ Member many-to-many relationship cleanly through RESERVATION. Unlike the hierarchical model, a record can participate in multiple owner-member sets.